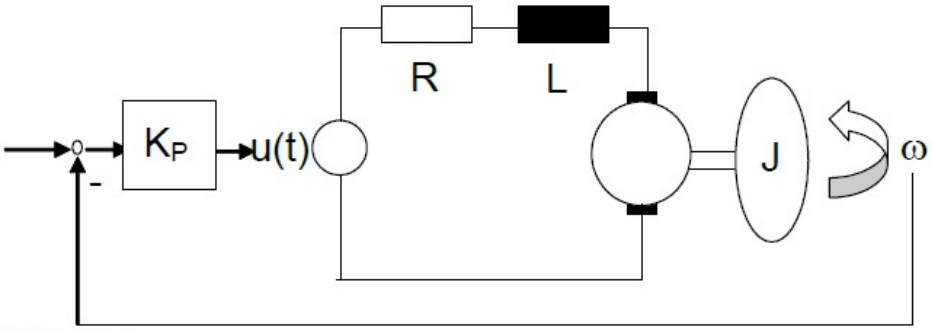
Exercise 4
Controller design for motor speed controller

Group 10
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Speed control is used in many application of mechatronic systems e. g. for robots. In this exercise a model of a dc-motor is given. Your task is to simulate the motor and than develop a controller using the Matlab tool-chain.



The equations are:

$$u(t) = R \cdot i(t) + L \cdot \frac{d}{dt}(i(t)) + c_m \cdot \dot{\alpha}(t); \quad \leftrightarrow \quad \frac{d}{dt}(i(t)) = \frac{1}{L}(u(t) - R \cdot i(t) - c_m \cdot \omega(t))$$

$$J \cdot \ddot{\alpha}(t) = c_m \cdot i(t) - D \cdot \dot{\alpha}(t); \qquad \frac{d}{dt}(\omega(t)) = \frac{1}{J}(c_m \cdot i(t) - D \cdot \omega(t))$$

```
p.L = 0.23e-3;  %[V/A*sec]      Induktivity
p.cm = 23.4e-3; %[Nm/A]         Motor-constant
p.R = 2.4;      %[Ohm]          Resistanace
p.J = 0.23e-6;  %[N*m^2]        Inertiay
p.D = 0.4191e-5;%[Nm/rad*sec]   Damping
```

Figure 1:

a) Simulate the motor-behavior for an input-voltage of 10 Volts.

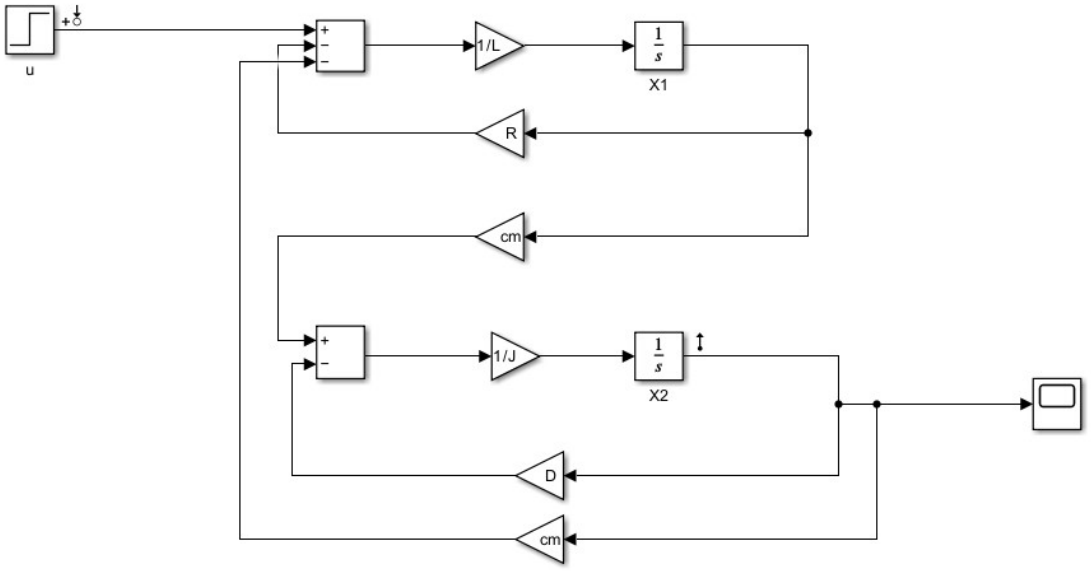


Figure 1:

```
    %% Motor Input with 10 Volts
clc;clf
L = 0.23e-3; %[V/A*sec] Induktivity
cm = 23.4e-3; %[Nm/A] Motor-constant
R = 2.4; %[Ohm] Resistanace
J = 0.23e-6; %[N*m^2] Inertiay
D = 0.4191e-5;%[Nm/rad*sec] Damping

out = sim('Excercise5a.slx');

plot(out.ex5.time, out.ex5.signals(1).values);
```

```
xlabel('Time [sec]');      % Label x-axis as time in seconds
ylabel('Speed '); % Label y-axis as speed in radians per second
ylabel('Speed [\omega]'); % Label y-axis as speed in radians per
   second with ohmega symbol
grid;                      % Enable grid for better visualization
title('Step Response of a DC Motor');
```

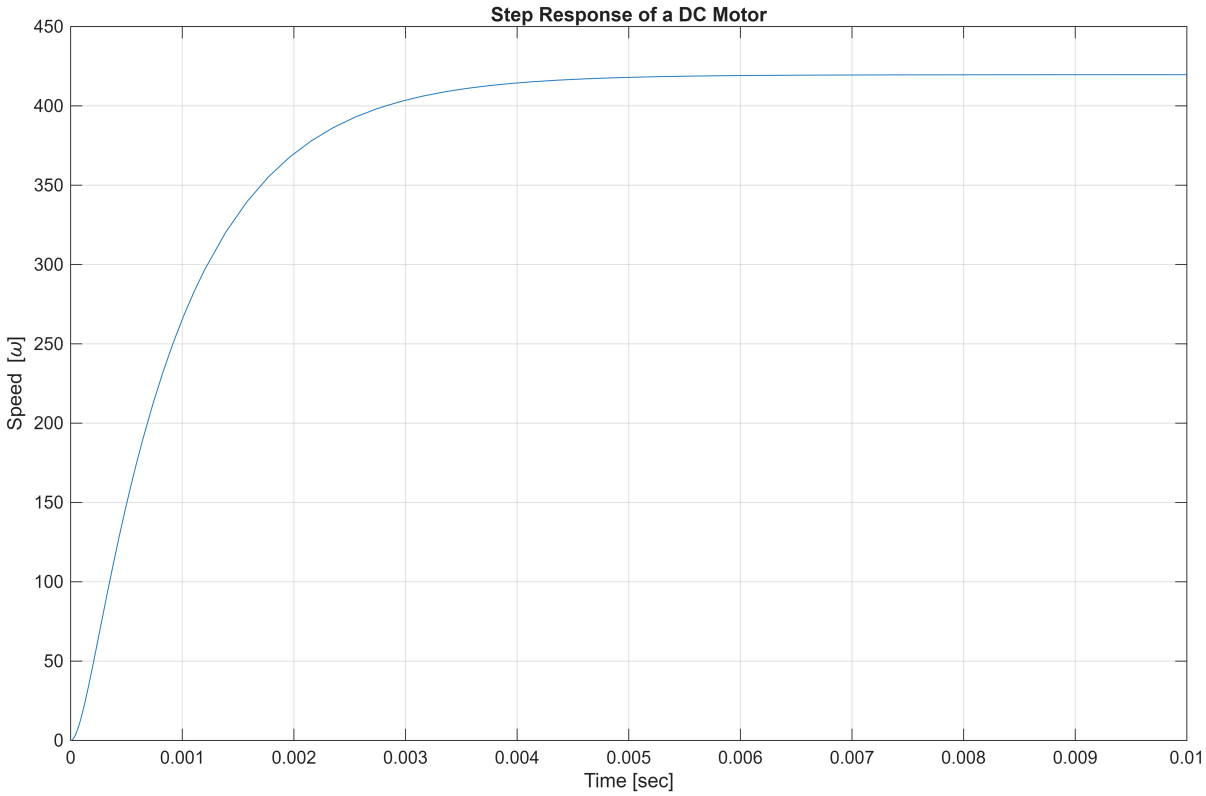


Figure 2:

b) Design a speed controller by using the Linearize-Tool and the Sisotool.of Matlab.

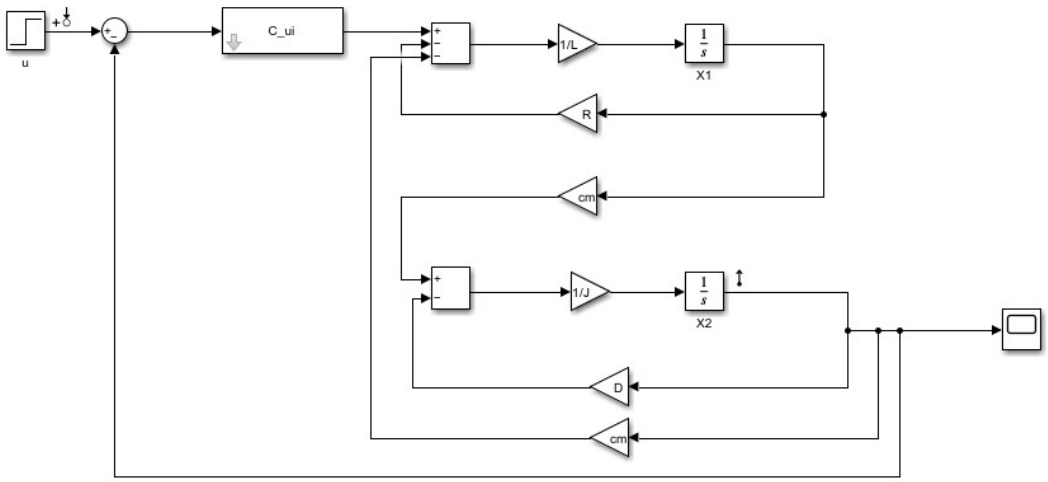


Figure 3:

A dynamic model of a DC motor was implemented in Simulink based on the electrical and mechanical differential equations of the system. The electrical subsystem consists of the armature resistance and inductance, while the mechanical subsystem is described by the motor inertia and viscous damping. The motor torque is proportional to the armature current via the motor constant, and the back-EMF is proportional to the motor speed.

A speed controller was designed using MATLAB's Linearize Tool and SISO Design Tool. The nonlinear Simulink model was linearized around a nominal operating point, and a proportional controller was tuned to achieve a fast and stable speed response with minimal overshoot and acceptable settling time. The controller output represents the motor input voltage and drives the electrical subsystem of the motor. The resulting closed-loop system shows stable behavior and accurately tracks the reference speed.

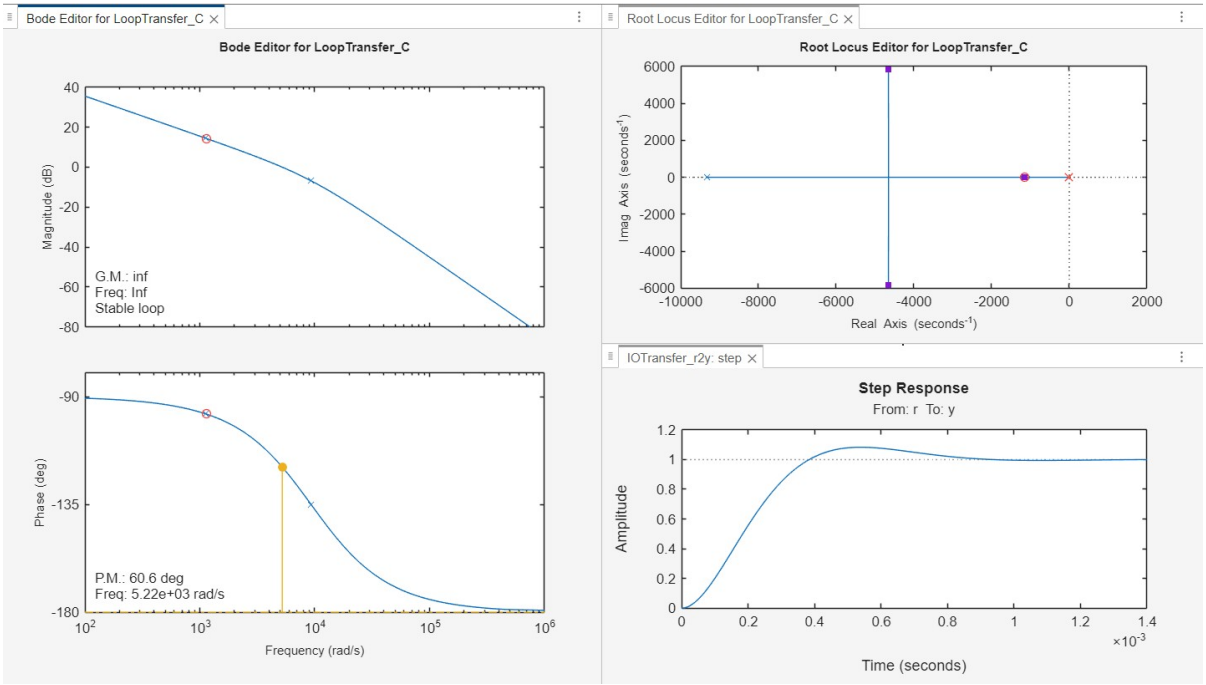


Figure 4:

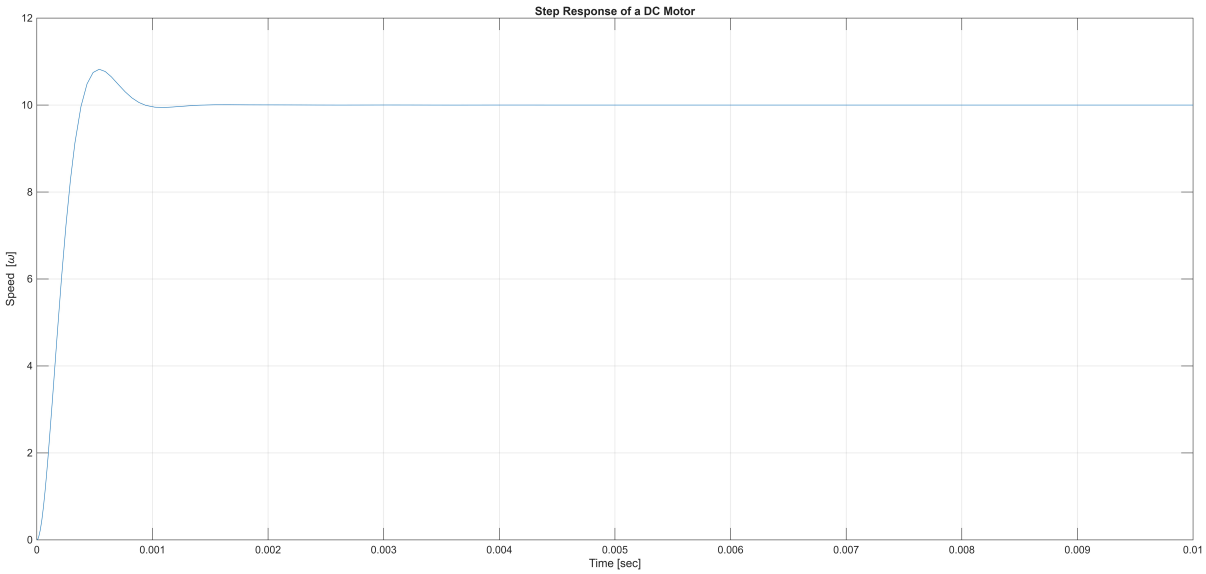


Figure 5:

c) Test the designed controller in Simulink and investigate the influence of voltage limits and quantizer-effects.

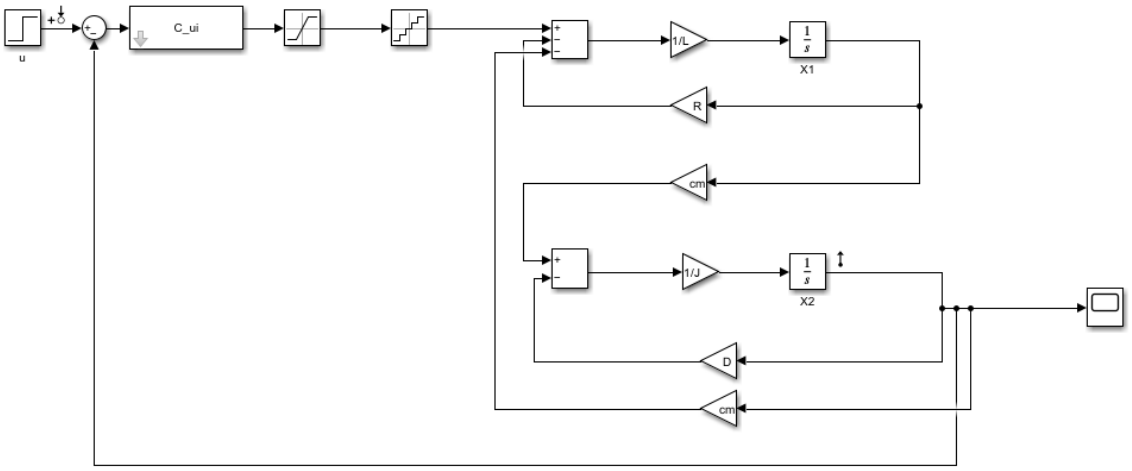


Figure 1:

To investigate actuator limitations and digital implementation effects, the designed controller was extended by including voltage saturation and quantization blocks in the Simulink model. A saturation block was placed directly after the controller output to model the physical voltage limits of the motor drive. This limits the input voltage to a defined maximum and minimum value and introduces nonlinear behavior into the closed-loop system.

Additionally, a quantizer block was inserted after the saturation block to represent finite digital-to-analog converter (DAC) resolution. The quantization results in a stair-stepped control voltage, which introduces small oscillations and ripple in the motor speed, particularly in steady-state operation. By comparing simulations with and without saturation and quantization, the influence of actuator constraints and digital effects on system performance could be clearly observed.

```
    %% Motor Input with 10 Volts
clc;clf
L = 0.23e-3; %[V/A*sec] Induktivity
cm = 23.4e-3; %[Nm/A] Motor-constant
R = 2.4; %[Ohm] Resistanace
J = 0.23e-6; %[N*m^2] Inertiay
D = 0.4191e-5;%[Nm/rad*sec] Damping

out = sim('Excercise5a.slx');

plot(out.ex5.time, out.ex5.signals(1).values);
xlabel('Time [sec]');      % Label x-axis as time in seconds
ylabel('Speed '); % Label y-axis as speed in radians per second
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   second with ohmega symbol
grid;                      % Enable grid for better visualization
title('Step Response of a DC Motor');
```

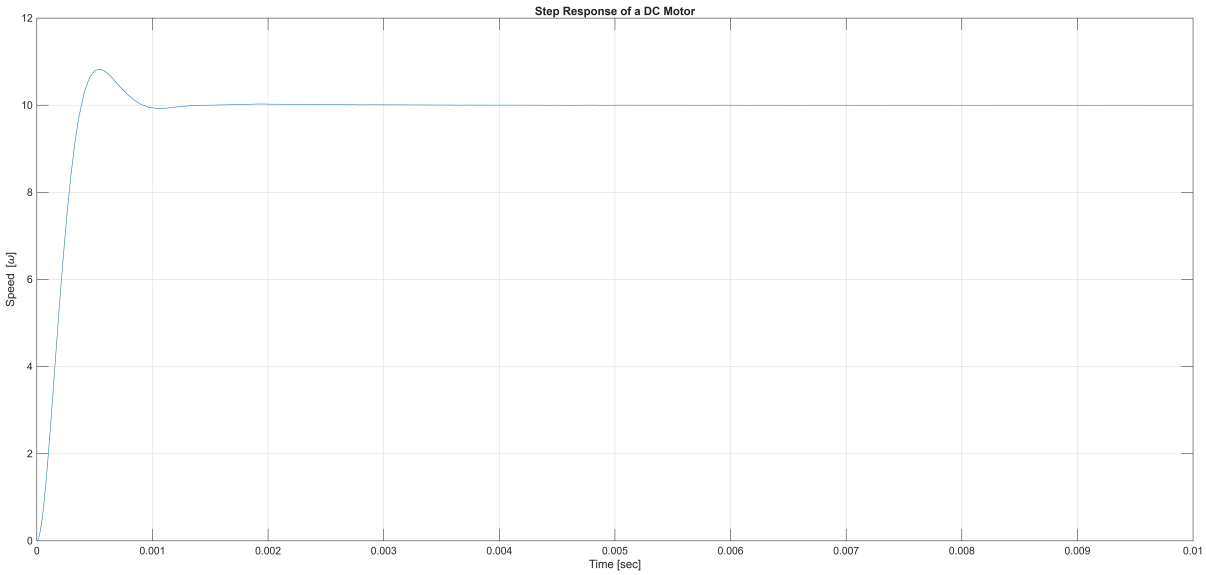


Figure 2: Stable Convergence (First Graph)This graph represents an ideally tuned or balanced system.Behavior: After the initial oscillation, the speed settles perfectly and remains flat at a constant value of 10.Meaning: The motor has reached its target speed, and there is no significant external interference or cumulative error.

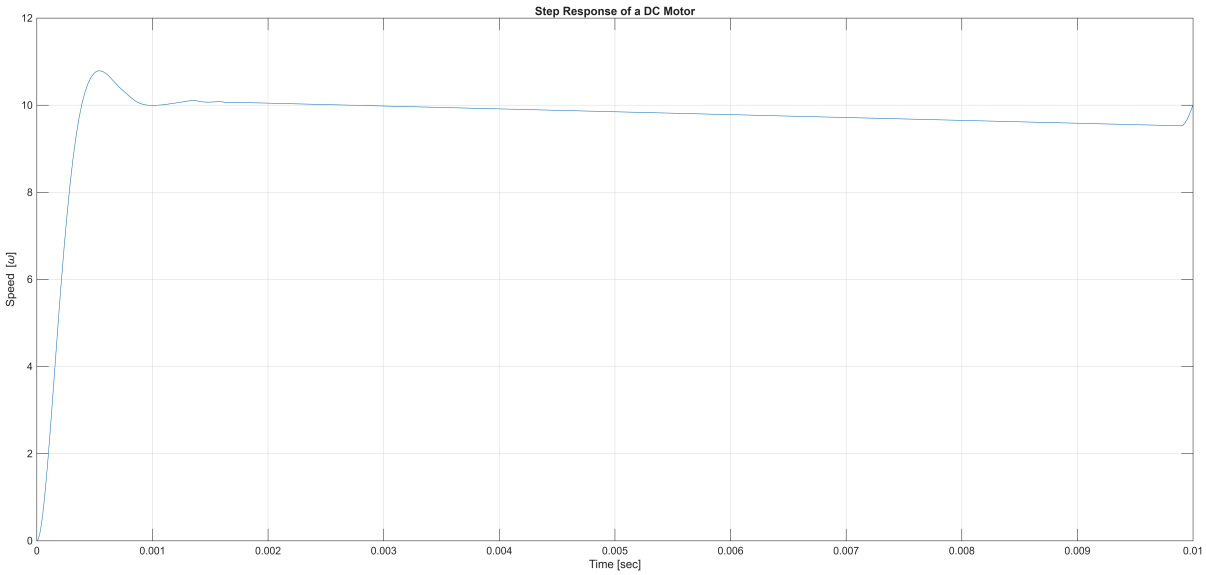


Figure 3: Negative Steady-State Drift (Third Graph)This graph shows the opposite of the second, featuring a steady decline.Behavior: After the initial peak, the speed gradually drops below the 10 mark, trending toward 9.5 before a small glitch/spike at the very end of the sample.Meaning: This could indicate increased loading/friction over time, a "droop" characteristic in the power supply, or a controller that is over-correcting and pulling the speed down.

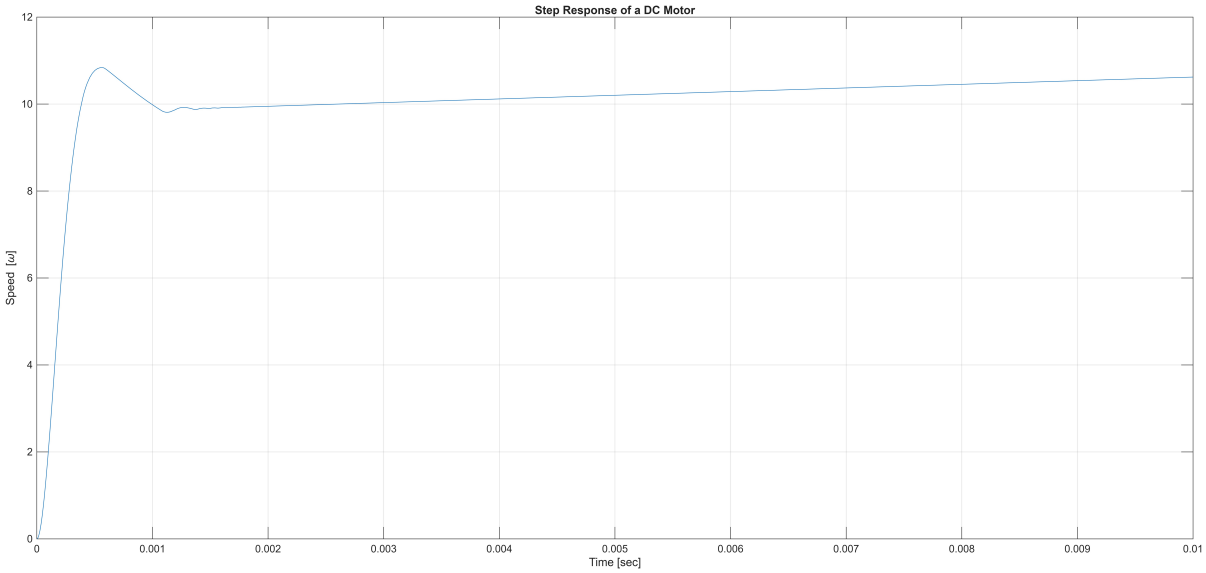


Figure 4: Positive Steady-State Drift (Second Graph)In this scenario, the system shows a ramp-up effect after the initial settling period.Behavior: Instead of staying flat at 10, the speed begins a slow, linear increase, ending near 10.6 by the 0.01s mark.Meaning: This often suggests an unstable integral component in a controller (like a PI or PID controller) or a system experiencing a constant reinforcing force/voltage increase.